Integrated Operations Platform

Technical and Commercial Proposal

Prepared for

D & S Electric Works

109, Industrial Area Extension, Rau, Indore 453331, Madhya Pradesh

Prepared by

PlotArmour

31/44 St 3, Block 17, Vishwas Nagar, Delhi 110032

GSTIN 07TQDPS9711L1ZT

+91 87440 69597

Date 20 October 2026 **Validity** 60 days

Status: DRAFT FOR FINAL REVIEW. Prices and scope are agreed; ownership and licence wording is subject to legal review before signature.

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Proposal: Verity for D & S Electric Works

Prepared for: D & S Electric Works (DSEW), 109, Industrial Area Extension, Rau, Indore 453331, Madhya Pradesh
Prepared by: PlotArmour, 31/44 St 3, Block 17, Vishwas Nagar, Delhi 110032, GSTIN 07TQDPS9711L1ZT, +91 87440 69597, team@plotarmour.in
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1. What we are offering

Verity is our own operations platform. We are offering DSEW a Verity solution configured for DSEW, **for a one-time fee**, in the options you asked for. On full payment DSEW receives the agreed deliverables and their source code, plus a perpetual licence to the pre-existing Verity components used within them. Everything is built inside Verity: no third-party software to buy.

We built the scope from your enquiry **and from our own understanding of how DSEW works**: an engineered, one-off product only stays trustworthy if what was approved is what was built, and nothing leaves the factory without its checks.

	Option 1: MVP	Option 2: Complete ERP
What it is	A usable core engineering and manufacturing ERP: enquiry to quotation to order, engineering and BOM, materials, manufacturing, travellers, quality	All 19 modules of your enquiry plus documented AI use cases, with deeper workflows and cross-module links
One-time licence and implementation fee (ex GST)	**₹3,25,000**	**₹4,80,000**
GST at 18% (illustrative; IGST expected as our GSTIN is registered in Delhi and DSEW is in Madhya Pradesh; confirmed at invoicing)	₹58,500	₹86,400
Total including GST	**₹3,83,500**	**₹5,66,400**
12-month bundled fee (section 7), ex GST	₹1,02,000	₹1,02,000
Contract value for the first 12 months, ex GST	**₹4,27,000**	**₹5,82,000**
Estimated delivery	About 6 to 8 weeks from kickoff	About 8 to 12 weeks from kickoff

Option 3: phased implementation. The Complete ERP delivered in four milestone-based phases, about 8 to 12 weeks from kickoff, each with its own price, demonstration and acceptance criteria (section 6 and document 2). It is an alternative delivery and payment structure for Option 2, at the same ₹4,80,000, not an additional purchase.

Timelines are estimates, not fixed deadlines. They assume the agreed scope, timely access to data, server, stakeholders and decisions, consolidated DSEW feedback, DSEW testing within the review windows, and agreed acceptance criteria. The final schedule is confirmed when requirements are finalised; delays from pending DSEW inputs, scope changes, third-party dependencies, additional requirements, or a hosting environment that fails the pre-deployment compatibility check (document 2, section 6) revise it by the corresponding time.

Use for either option: DSEW's own business, one site, up to 15 named users. No annual or per-user licence fee; the only recurring charge is the 12-month bundled fee. More users, sites or new modules are priced separately, and any extra feature needs a written change request with a revised cost or schedule. Prices are fixed and not discounted further.

Scope in one line. The MVP is roughly 55% of the Complete ERP's functional scope (a planning estimate, not a measured figure). It is priced at about 68% of it, because the foundations (architecture, sign-in, permissions, numbering, deployment, testing, documentation) are needed in both.

2. Included in both options

Responsive interface with searchable, filterable tables; role-based permissions; audit trail; configurable numbering for the agreed document types; backup and a tested restore; installation on one DSEW Linux VM; testing and UAT; training; administrator and user guides and runbooks; a documented export of DSEW's data (records as CSV, files in original form with an index); and the 12-month bundled fee terms in section 7.

3. Option 1: MVP, Core Engineering and Manufacturing ERP

One connected workflow: **customer enquiry, quotation and order confirmation, project, engineering documents and BOM, material availability and purchasing, manufacturing job and traveller, quality inspection and release, delivery readiness.** Creating a manufacturing job identifies its BOM, shows material availability, records material issues, tracks operations and the traveller, and blocks final release while mandatory quality checks are incomplete.

#	Module / deliverable	Price (₹, ex GST)
1	CRM, enquiry and quotation management	35,000
2	Project and order management	25,000
3	Document management and configurable numbering	35,000
4	Basic engineering and BOM management	35,000
5	Basic procurement and inventory management	40,000
6	Manufacturing execution and job management	40,000
7	Manufacturing traveller management	25,000
8	Core quality management	30,000
9	Operational dashboards and basic notifications	15,000
	Module development subtotal	2,80,000

#	Module / deliverable	Price (₹, ex GST)
	Requirements finalisation and workflow design	12,000
	Application configuration and deployment	10,000
	Integrated testing and UAT support	12,000
	Documentation, training and handover	11,000
	Implementation subtotal	45,000
	MVP total (ex GST)	**3,25,000**

MVP boundary. Basic BOMs, material records, purchasing, stock balances and job-level material allocation are included. Excluded: advanced MRP, barcode-driven genealogy, finance and accounting, payroll and HR, dispatch workflow, contracts and compliance tracking, AI, and advanced enterprise workflows.

4. Option 2: Complete Integrated ERP

#	Module / deliverable	Price (₹, ex GST)
1	CRM, enquiry, quotation and sales order management	30,000
2	Advanced project management	20,000
3	Engineering and product lifecycle management	30,000
4	Document management and controlled document workflows	25,000
5	Manufacturing execution	30,000
6	Manufacturing travellers and process documentation	15,000
7	Quality management system	25,000
8	Barcode, QR code and traceability	15,000
9	Procurement and vendor management	20,000
10	Inventory and warehouse management	20,000
11	Production planning and basic MRP	20,000
12	Testing and equipment management	15,000
13	Finance and accounting, built inside Verity	20,000
14	HR and payroll management	15,000
15	Dispatch and logistics	10,000
16	Contracts, compliance and risk	10,000
17	Management dashboards and business intelligence	15,000
18	Notifications and workflow automation	15,000
19	AI, documented use cases only (section 9)	15,000
	Module development subtotal	3,65,000

#	Module / deliverable	Price (₹, ex GST)
	Requirements analysis and solution architecture	25,000
	UI/UX design and workflow configuration	20,000
	Cross-module integration and deployment	25,000
	Integrated testing and UAT support	20,000
	Documentation, training and knowledge transfer	15,000
	Implementation subtotal	1,05,000
	Cross-module dependency coordination and project governance	10,000
	Complete ERP total (ex GST)	**4,80,000**

Complete boundary. All 19 modules and the documented AI use cases at the **agreed baseline depth**: standard workflows and limited customisation. The detailed specification and module acceptance criteria fix that baseline before development starts. It is not an unlimited implementation of every advanced capability in the enquiry. Explicitly outside it: advanced AI, bulk historical data migration, statutory compliance certification, and any integration with third-party systems.

Price difference: ₹1,55,000, so Option 2 costs about 47.7% more than Option 1 for the extra breadth.

5. MVP versus Complete, module by module

Module	Option 1: MVP	Option 2: Complete
CRM	Customer records, enquiries, follow-ups	Full pipeline, RFQ and tender tracking, correspondence, complaints, advanced history
Project management	Projects, milestones, tasks, deadlines, status	Dependencies, WBS, budgets, risk and change management, advanced reporting
Enquiry, quotation, sales	Quotations, basic costing, revisions, order confirmation	Advanced costing, alternative offers, PO comparison, sales orders, retention, LDs, milestone billing
Engineering and PLM	Product and component records, BOMs, drawing attachments, revisions	Multi-level BOMs, engineering change orders, formal design approvals
Document management	Repository, numbering, revision history, search, basic approvals	Full controlled-document lifecycle, transmittals, acknowledgements, advanced audit and distribution
Manufacturing	Job cards, operation sequences, material allocation, status	Advanced routing, work-centre scheduling, labour tracking, WIP, subcontracting, bottleneck analytics
Travellers	Configurable travellers, checkpoints, sign-offs	BOM-driven travellers, QR codes, full genealogy, customer-facing formats
Quality	Incoming, in-process and final inspection, NCRs, quality release	QAP/ITP, CAPA, audits, calibration, supplier quality, analytics
Barcode, QR, traceability	Basic material and job references, manual tracking	Barcode scanning, batch, heat and serial tracking, product genealogy

Module	Option 1: MVP	Option 2: Complete
Procurement	Suppliers, purchase requests, receipts, purchasing	Approval workflows, RFQs, comparative statements, supplier performance, subcontracting
Inventory	Stock balances, material issues, reservations, shortages	Multi-warehouse and bins, valuation, reconciliation, ageing, advanced traceability
Planning and MRP	Job priorities, manual material planning	Automated BOM-based MRP, capacity planning, scheduling, shortage recommendations
Testing and equipment	Basic test records and attachments	Dedicated test workflows, automated reports, calibration, preventive maintenance
Finance and accounting	None	Invoicing and payment status, GST support, reconciliation, receivables and payables, profitability, all inside Verity
HR and payroll	None	Employees, attendance, leave, payroll, statutory calculations, competency tracking
Dispatch	Delivery readiness tracking	Dispatch planning, packing lists, challans, shipment tracking, partial deliveries
Contracts and risk	Basic project and order references	Obligations, bank guarantees, LD exposure, insurance and NDA tracking, risk registers
Dashboards	Operational dashboards for the included workflows	Cross-department KPIs, profitability, ageing, cash flow, supplier and resource analytics
Notifications	Basic reminders and alerts	Configurable workflow engine, escalations, cross-module triggers, automated documents
AI	Not included	Documented use cases only (section 9)

Boundaries and acceptance for the higher-risk modules (Option 2 and Option 3)

The detailed specification fixes each module's depth in writing before development starts. These are the boundaries we will hold to.

Area	Included at the agreed depth	Excluded	Accepted when
Finance and accounting	Invoice records from orders and milestones, recorded payments and receipts, receivables and payables views, configurable GST rates, job profitability	GST return and portal filing, e-invoicing, bank feeds, statutory audit certification, multi-company consolidation	An invoice from an order shows tax and outstanding amount; a recorded payment reduces it; receivables reconcile to invoices and payments; job profitability reconciles to recorded costs
HR and payroll	Employee records, attendance, leave, shifts, timesheets, salary structures, payroll runs using statutory components configured from DSEW-supplied rules, payslips	Statutory filing and portal submission, legal compliance certification, attendance-device integration, rule changes DSEW has not configured	A sample month for agreed employees reproduces DSEW's own calculation using its supplied rules

Area	Included at the agreed depth	Excluded	Accepted when
AI	Only use cases written into the specification, each with defined inputs, outputs and an acceptance threshold on agreed sample documents	Custom model training, autonomous actions, unbounded usage, any approval or release by AI	Each documented use case meets its threshold on the agreed samples
Engineering and revisions	Product and component masters, multi-level BOMs, revisions with release, engineering change requests and orders with approval, drawing attachments	CAD integration or editing, simulation tools, 3D viewing	A released revision cannot be edited; an order records its revision; a change creates a new revision through approval
Quality workflows	Inspection plans, checklists and acceptance criteria, NCR and CAPA, calibration records, QAP/ITP templates for the agreed formats, approvals inside the system	Customer formats beyond the agreed number, drafting of quality manuals, accreditation or certification work	A failed or missing mandatory check blocks release; an NCR closes only after CAPA verification
Integrations	Everything is inside Verity; e-mail notifications through DSEW's mail server details	Any third-party or external system, WhatsApp, e-way bill, bank, GST portal, attendance devices; barcode scanner and printer hardware	Not applicable; added only by written change request

The MVP includes only the basic engineering, procurement, inventory and quality layers described in sections 3 and 5.

6. Option 3: the Complete ERP in four phases, about 8 to 12 weeks

Phase	Content	Price (₹, ex GST)
1. Foundation and commercial core	Requirements and architecture, sign-in and roles, CRM, quotation and sales order, projects, document numbering, initial engineering masters and BOM	1,20,000
2. Supply chain and manufacturing core	Engineering revisions and release, procurement, inventory and material allocation, planning and basic MRP, manufacturing jobs and operations	1,35,000
3. Quality, traceability and operational controls	Travellers, quality and NCRs, barcode/QR and traceability, testing and equipment, dispatch, contracts and compliance	1,30,000
4. Business modules, validation and go-live	Finance and accounting, HR and payroll, dashboards, notifications and automation, documented AI use cases, end-to-end testing, UAT, deployment, training, handover	95,000
Total	**About 8 to 12 weeks**	**4,80,000**

Phase prices allocate the Option 2 total; payments follow the milestones in section 8. The calendar for each phase is flexible across the 8 to 12-week window and, with each phase's acceptance criteria, is in document 2.

7. The 12-month period: warranty, testing support and maintenance

DSEW will test Verity in its real workflows for **12 months from go-live**. For that whole period we charge one bundled fee of **₹8,500 per month** (₹1,02,000 ex GST for 12 months; ₹10,030 per month including 18% GST), for either option. **Start date:** the go-live date, when the go-live gates have passed and the handover is signed, not project kickoff. **Billing:** 12 monthly invoices, each issued in advance on the go-live date and then on the same date each month, payable within 7 calendar days, with GST added.

The bundled fee replaces a separate warranty, a separate testing-support charge and a separate maintenance contract. None of them is charged on top during the 12 months. It covers exactly three things:

1. **Defect warranty.** Correcting any reproducible defect where Verity does not meet the signed requirements, whether found in UAT or at any time in the 12 months. A defect caused by DSEW's own changes, hardware, network or third-party software is not a Verity defect.
2. **Testing and UAT assistance.** Helping DSEW run the agreed UAT scripts and test real jobs, reproducing and triaging each reported issue, recording DSEW's feedback, and reviewing reported issues and feedback with DSEW at an interval agreed at the start.
3. **Routine maintenance.** Keeping the DSEW installation on Verity's tested releases (stability and security updates), helping run and check backups and restores, and administrator guidance.

Service terms. Support is remote, by email (team@plotarmour.in) and phone (+91 87440 69597), on business days (Monday to Saturday, 10:00 to 18:00 IST). The fee includes up to 8 hours a month of testing assistance and maintenance; more is quoted separately. **Response targets:** we acknowledge a report within 1 business day; a reproducible defect is given a fix or workaround plan within 3 business days, with the fix time depending on the defect. **Escalation:** Shubhankar Rakshit, CFO (team@plotarmour.in, +91 87440 69597).

Not covered. Each is quoted separately and needs DSEW's written approval before any work: new features; changes to agreed workflows or to a requirement DSEW has already accepted; additional integrations; new product families, reports, formats, users or sites; data cleansing or migration beyond the agreed imports; hardware or network faults; and on-site work and travel, which are chargeable separately and need prior approval. DSEW's testing feedback is used to prioritise and quote these.

After the 12 months, no warranty, testing support or maintenance is included. Any support afterwards needs a separately agreed renewal in writing. No annual contract is part of this offer.

Remote first: delivery and support are remote; the fee assumes no on-site work.

8. Payment

Each invoice is tied to its milestone, carries GST at the rate then applicable, and is **payable within 7 calendar days**. The same structure applies to Option 1, Option 2 and Option 3.

Milestone	Share	Option 1 ₹ ex GST	Option 2 / 3 ₹ ex GST
1. Kickoff: signed order, product owner named, requirements signed off	30%	97,500	1,44,000
2. Core workflow demonstration accepted: the commercial core and first working release	30%	97,500	1,44,000

Milestone	Share	Option 1 ₹ ex GST	Option 2 / 3 ₹ ex GST
3. Integrated testing and UAT build delivered to DSEW	30%	97,500	1,44,000
4. Handover and go-live: UAT signed off, go-live gates passed, handover signed	10%	32,500	48,000
Total	**100%**	**3,25,000**	**4,80,000**

Review window. DSEW has **5 business days** after each milestone delivery to report material deviations from the agreed criteria. In-scope defects are corrected before sign-off; new requirements follow change control. No report in the window means the milestone is accepted, so approval is never open-ended. Each milestone has written deliverables and acceptance criteria in the specification. The 12-month bundled fee is invoiced separately (section 7).

9. Artificial intelligence (Option 2 and 3 only, documented use cases)

Options 2 and 3 include only the AI use cases written into the specification, each with defined inputs, outputs and an acceptance threshold on agreed sample documents. AI is never an approver and always works inside each user's permissions. It does not approve quotations, release documents, alter records, authorise quality releases or act autonomously, and no essential workflow depends on it. Custom model training, unbounded AI usage and advanced bespoke models are excluded unless separately agreed. The table below shows which of the capabilities you listed can run locally and which would need an external service; the specification selects which are delivered.

AI capability you listed	Local on DSEW's server	External service
Natural-language search across authorised documents and records, including past quotations, POs, BOMs and jobs	Feasible with open models and search	Not needed
Retrieval from specifications, QAPs, travellers and project history	Feasible	Not needed
Document classification and metadata extraction	Feasible, accuracy to be tested	Optional for hard cases
Missing-document and incomplete-record detection	Feasible (mostly rules)	Not needed
Natural-language questions over operational data	Feasible, read-only	Optional
Drafting quotations, estimates, work instructions, inspection plans and reports from approved templates	Feasible with smaller models; quality limited	Optional for better quality; usage charges
Management summaries	Feasible	Optional

Local models need suitable hardware (a GPU workstation is the usual choice) and carry no usage charges. External services are optional, charged by their providers, and off unless DSEW enables them. Their charges and any local GPU hardware are not in the fee.

10. Commercial headings you asked for

#	Heading	Position
1	Requirement analysis and system design	Included (Option 1: ₹12,000; Option 2: ₹25,000)
2	UI/UX design and prototyping	Included (Option 2: ₹20,000; Option 1: mock-ups of printed formats and key screens inside the workflow design line)
3	Development charges by module	The module tables in sections 3 and 4
4	Third-party software and licences	None
5	AI model and API costs	Documented use cases in Options 2 and 3 only; external service charges and GPU hardware not included
7	Data migration and initial configuration	Initial master-data setup included (CSV import of agreed masters); cleansing and bulk historical migration excluded
8	Testing and deployment	Included in the implementation lines
9	Training and documentation	Included in the implementation lines
10	Server, infrastructure and cybersecurity	DSEW's own costs, estimated in document 2; security measures in document 2; independent penetration testing excluded
11	Annual maintenance contract	None. The first 12 months are covered by the bundled fee; later support is a renewal agreed in writing
12	Warranty and post-deployment support	The bundled fee in section 7
13	Change-request rates	Each request is quoted separately with a revised cost or schedule and approved in writing; no rate card is published. On-site work and travel are chargeable separately with prior approval
14	Recurring per-user, per-module or subscription fees	None, apart from the ₹8,500 a month for the first 12 months
15	Taxes	GST at 18% on all fees (see section 1)

Cost over three to five years. Fixed today: the option price plus ₹1,02,000 in the first 12 months (₹4,27,000 or ₹5,82,000 ex GST). Not yet priced: support after 12 months and any later enhancements. DSEW's own infrastructure is estimated in document 2. No third-party licence fees apply.

11. Ownership and licence in short

DSEW owns its business data and can export it. **On full payment DSEW receives the agreed deliverables and their source code.** PlotArmour keeps ownership of its pre-existing tools, reusable libraries, frameworks and general know-how (including Verity's core) and gives DSEW a perpetual licence to use them within the delivered solution. The wording is to be reviewed by counsel; open points are listed in document 3.

12. What we have and have not verified

Verity's foundations (data isolation, permissions, audit, stock, manufacturing orders, QC checkpoints, project jobs, approvals, QR identity) exist and have been run on our test environments with fictional data. Revision-controlled BOMs, controlled documents, structured test records, HR and payroll, finance and the other modules beyond the foundations are built during the project, which is why the timelines are ranges. We have **not** yet tested on DSEW's host or on Windows Server 2019 (a pre-deployment compatibility check comes first, document 2), run a real-phone HTTPS upload, rehearsed the combined database, files and identity restore, or run the outage tests. These are go-live gates (document 2); **production use is blocked until they pass.**

Delivery, Deployment and What We Need

D & S Electric Works. How the work is delivered, how it runs on your premises, and what we need from you.

> **Status: DRAFT FOR FINAL REVIEW.**

1. Estimated timelines

Option	Estimated implementation timeline
Option 1: MVP ERP	Approximately **6 to 8 weeks** from project kickoff
Option 2: Complete ERP	Approximately **8 to 12 weeks** from project kickoff
Option 3: Phased Complete ERP	Approximately **8 to 12 weeks**, delivered through four phases with defined milestones and acceptance criteria

These are estimates, not fixed deadlines. They assume the agreed scope, timely access to the data, server, stakeholders and decisions we need, consolidated DSEW feedback, DSEW testing within the review windows, and agreed acceptance criteria. **The final schedule is confirmed when requirements are finalised.** Delays caused by pending DSEW inputs, scope changes, third-party dependencies, additional requirements, or a hosting environment that fails the pre-deployment compatibility check (section 6) revise the schedule by the corresponding time.

Kickoff is the latest of: signed order; product owner named; requirements inputs delivered (sample documents, numbering conventions, server survey answers, access).

2. Option 3: the four phases

Each phase ends in a working demonstration and a written review. The phase prices are unchanged by the timeline.

Phase	Weeks (8-week pace / 12-week pace)	Price (₹, ex GST)	Delivered and accepted when
1. Foundation and commercial core	1-2 / 1-3	1,20,000	DSEW can create linked customer, enquiry, quotation, sales order and project records with controlled document numbering, and initial engineering masters and BOM
2. Supply chain and manufacturing core	3-4 / 4-6	1,35,000	A BOM drives material requirements and availability for a manufacturing job; purchases, receipts and stock movements are recorded; engineering release works
3. Quality, traceability and operational controls	5-6 / 7-9	1,30,000	Operations, travellers and quality inspections are linked; agreed traceability, testing, dispatch and contract controls work; failed mandatory checks block release
4. Business modules, validation and go-live	7-8 / 10-12	95,000	Finance, HR and payroll, dashboards, notifications and the documented AI use cases work; end-to-end UAT passes; deployed, trained and handed over

Phase	Weeks (8-week pace / 12-week pace)	Price (₹, ex GST)	Delivered and accepted when
Total	**8 to 12 weeks**	**4,80,000**	

Stage	What happens
Kickoff	Discovery and architecture: workflows, data model, roles, mock-ups of printed formats and key screens, written module depth and acceptance criteria. **First install on the DSEW VM, HTTPS access, real-phone upload test**
Phase 1 demonstration	Commercial core
Phase 2 demonstration	BOM to material to job, production status, basic MRP. **First backup and restore rehearsal**
Phase 3 demonstration	Travellers, quality, traceability, testing, dispatch, compliance
Phase 4 and go-live	Business modules and AI use cases, full UAT, **final restore test**, deployment, training, handover

3. Option 1: the MVP stages

Stage	Estimated weeks	Accepted when
Kickoff and specification	1	Requirements and module depth signed off; first install on the DSEW VM
Commercial core	2-4	Customer, enquiry, quotation, order, project and numbered documents linked
Engineering, materials and manufacturing	3-6	BOM, purchasing, stock and issue to a manufacturing job; traveller and operations tracked
Quality, dashboards and go-live	5-8	Inspections and quality release gate work; dashboards and notifications; UAT passes; deployed, trained and handed over

4. How we work

- **Specification and design:** produces a signed specification (workflows, roles, numbering, templates, formats, the agreed depth of every module and acceptance criteria), an architecture and database overview, and approved mock-ups.
- **Milestones and acceptance:** DSEW has **5 business days** after each demonstration or delivery to accept it or report material deviations from the agreed criteria. In-scope defects are corrected and shown again before sign-off; new requirements follow change control. No report in that period means accepted. Payment milestones are in document 1, section 8.
- **Issue tracking:** a shared issue list visible to DSEW from day one, with severity, owner and status. Critical (wrong data, security issue, or work blocked with no workaround) and High (major function impaired) issues block go-live.

- **UAT:** scripts agreed in advance (enquiry to quotation to order; BOM release; material to job; failed test blocking dispatch; dispatch released after approval; data export; backup and restore). Each scenario is signed off with date, tester and result.
- **Definition of done:** server-side authorisation, tenant-isolation test, audit entry, validation and error states, automated tests, documentation note, demonstrated in the review.
- **Dependencies between modules:** numbering and approval precede BOM release; a released BOM precedes material planning, manufacturing orders and travellers; test templates precede dispatch gating; stock and invoicing data precede finance and dashboards.
- **Changes:** anything beyond the agreed scope needs a written change request with a revised cost or schedule, approved before work.
- **Remote first:** delivery is remote. On-site work and travel are chargeable separately and need prior approval.

5. Environments

- **Company-hosted test environment:** demonstrations and early hands-on use.
- **One DSEW Linux VM:** installed at kickoff, used for UAT, and used as production only after the gates pass. A second production VM is not included. If the surveyed host cannot support this, a different arrangement is quoted separately.

6. Go-live gates (none completed yet)

1. Server survey complete and the DSEW host confirmed suitable.
2. Installation and HTTPS access work on the DSEW host, including a real-phone upload.
3. **Combined backup and restore** of database, files and identity configuration onto a clean environment passes.
4. Identity-provider and object-store outage behaviour tested: existing sessions keep working, new sign-ins or uploads fail clearly, quality gates stay enforced.
5. UAT scripts pass.
6. The data export reconciles with the system.

Go-live happens when these gates pass and the handover is signed. The 12-month support period starts on that date.

Pre-deployment compatibility check

The hosting environment has not been tested. Before anything is installed on DSEW's host, we run a non-intrusive check in the first days after kickoff and record pass or fail for each dependency. Nothing on the host is changed during it.

Dependency	What we need	DSEW provides
Host and virtualisation	Hyper-V role enabled (or a separate small host), hardware virtualisation switched on, room for one VM of 4 vCPU, 16 GB RAM and 500 GB (the Complete ERP and AI may need more)	Host access; licence confirmation from your Microsoft partner
Guest operating system	An Ubuntu Server LTS VM runs under that host	Installation media access

Dependency	What we need	DSEW provides
Network	A fixed address and internal DNS name for the VM; gigabit LAN; shop-floor phones and PCs reach it over HTTPS	DNS entry, firewall rules
TLS certificate	A certificate for the internal name, trusted by office PCs and phones	Certificate or internal certificate authority
Remote access	DSEW's VPN, if remote use is wanted	VPN access
Internet for install and updates	Controlled access, or offline media	Outbound-network policy decision
Backup and power	A separate backup target at least equal to the data size; UPS	Storage and UPS
Administration	A named IT contact with the rights needed during the check	Contact and access

Outcome. On a pass, installation proceeds. On a fail, we give a written report and an alternative (a second VM, a dedicated small host, or a different runtime) quoted separately, and the schedule is revised accordingly.

7. Running on Windows Server 2019

Verity runs on Linux. We do **not** propose running it natively on Windows Server 2019, and we do not recommend Docker Desktop or WSL for production there. The plan is one Ubuntu Server VM under the Hyper-V role on your server, or a small dedicated host. **This has not been tested on your server.** Licensing of the Windows host is for your Microsoft partner.

8. Your 15 questions

#	Question	Answer
1	Architecture and stack	One web application (TypeScript, Next.js, React), PostgreSQL 16 database, Keycloak sign-in, S3-compatible file storage, Docker. All open source, no licence fee
2	Direct on Windows Server 2019?	No: Linux VM under Hyper-V or a small separate host
3	Hardware	For 15 users: 4 vCPU, 16 GB RAM, 500 GB mirrored SSD, separate backup target, UPS. Confirmed after survey; the Complete ERP and AI may need more
4	Database licence	PostgreSQL, no fee
5	Browsers and devices	Current Chrome, Edge, Firefox; Android and iOS phones and tablets for the shop floor
6	Network, remote access	LAN; remote only through your VPN
7	Backup and recovery	Scripts exist; combined restore is a go-live gate. Recovery time and data-loss window agreed with you
8	Security, authentication, audit	TLS, role-based permissions, per-company data isolation in the database, audit trail the application cannot change or delete, MFA available in Keycloak and enabled if you ask
9	Encryption, secure remote access	TLS in transit; encrypted backups and disk encryption recommended, confirmed at acceptance; VPN for remote use

#	Question	Answer
10	Performance and document growth	Designed for 15 named users; load test is part of acceptance; documents on a separate volume sized with you
11	Offline	In-LAN use needs no internet after install; installs and upgrades need controlled internet or offline media
12	Integration, API	Permissioned REST API. Everything in the offered scope is inside Verity; no external integration is included
13	Source code, repository, version control	Source code of the DSEW-specific deliverables is handed over on full payment, in a repository DSEW controls (document 3)
14	Test, staging, production	Section 5
15	Upgrades, maintenance, long-term support	Versioned releases with tested upgrade and rollback; covered during the 12-month period; support after it is a separately agreed renewal

Modular expansion: new departments, workflows, product families and users are added by configuration or by adding modules to Verity, without redeveloping what is already there.

9. User interface

A responsive interface for office PCs and shop-floor phones and tablets; searchable, sortable, filterable lists; direct links between related records (a project links to its customer, quotation, order, BOM, job, traveller, tests and documents); changes to approved records follow the revision or amendment path; role-based permissions; printable branded documents. English in this scope; Hindi interface labels are not included.

10. What is handed over

On full payment: the working, tested application; the modules of the chosen option; the source code of the DSEW-specific deliverables; the architecture overview and database schema for the delivered scope; deployment and configuration instructions; backup and recovery procedures; administrator and user guides; document templates and workflow configuration register; API notes for what the scope uses; test reports and UAT sign-off records; training; administrator credentials through a signed handover checklist; and the 12-month terms in document 1.

11. Assumptions and exclusions

Assumed: up to 15 named users, one site, English only; DSEW supplies samples, decisions and a numbering convention; DSEW provides the host, TLS certificates, internal DNS, VPN access, backup storage and a network contact.

Excluded: anything outside the chosen option's module tables; historical transaction migration; hardware and networking; GST on fees; on-site work and travel; support after the 12-month period; a second VM; independent penetration testing; Hindi interface; attendance-device and e-way bill integration; external AI service charges and AI hardware; DSEW staff time.

DSEW's own costs (planning estimates, not quotations): backup target and UPS ₹0.8 to 1.2 lakh; a new server only if the existing one cannot run the VM, ₹3.5 to 4.5 lakh; platform software licences: none beyond this proposal.

12. What we need from you

1. A named product owner (about 3 to 4 hours a week) and leads for quality, engineering, stores and finance.
2. A walk-through of one recent job from enquiry to dispatch, and your numbering convention.
3. Who drafts, reviews, approves and releases drawings, BOMs and test records; is self-approval ever allowed?
4. Samples: drawing, released BOM, quotation, traveller, test report or certificate, packing list, delivery challan, inspection formats and document templates.
5. The tests per product family and acceptance limits; how instruments and calibration are tracked.
6. Server make, OS build, CPU, RAM, free storage, Hyper-V status, network and VPN set-up, backup destination, IT owner, recovery time and data-loss window you accept, outbound-network policy, UPS.
7. Agreement to the licence and ownership terms in document 3 before the order.

Licence and Ownership

D & S Electric Works. Position on section 5 of your enquiry.

> **Status: DRAFT FOR FINAL REVIEW.** The position in section 1 is the company's decision. The final wording must be reviewed by counsel before it is relied on, and nothing here is a contract until signed.

1. The position

- **DSEW owns its business data**, including everything it enters or uploads, and can take a documented export (records as CSV, files in original form with an index) at handover and once more on request during the 12-month period.
- **On full payment, DSEW receives the agreed deliverables and their source code:** the DSEW-specific code, configuration, templates and documentation produced for this project, handed over in a repository DSEW controls.
- **PlotArmour keeps its pre-existing assets:** Verity's platform, tools, reusable libraries, frameworks and general know-how stay PlotArmour's. **DSEW receives a perpetual, non-exclusive licence to use those components within the delivered solution** for its own business, at one site, for up to 15 named users, included in the one-time fee. More users or sites are priced separately.
- We do not use DSEW's confidential drawings, prices, specifications or quality formats for anyone else, under a mutual NDA.
- Open-source components used (for example PostgreSQL, Keycloak, Next.js) stay under their own licences, listed in a licence inventory at handover. No paid third-party component is needed.

2. Your nine questions

#	Question	Answer
1	Complete source code delivered?	The source code of the DSEW-specific deliverables, on full payment. Verity's pre-existing platform components are licensed for use within the solution; source access to them is an open point (section 3)
2	Does DSEW own the custom code and IP?	DSEW owns its business data and receives the agreed deliverables and their source code on full payment. PlotArmour retains pre-existing tools, libraries, frameworks and know-how. The exact ownership-versus-licence wording for the DSEW-specific code is for counsel (section 3)
3	Can DSEW or another developer modify the code?	Yes for the DSEW-specific deliverables handed over. For pre-existing components, subject to section 3
4	Can it run without dependence on you?	It runs on DSEW's server day to day, with runbooks for install, backup, restore and upgrade. Updates to Verity's pre-existing components come from PlotArmour
5	Proprietary frameworks, paid components, recurring licences?	PlotArmour's pre-existing components are licensed as above. No paid third-party components or recurring licence fees; the only recurring charge is the 12-month bundled fee in document 1

#	Question	Answer
6	Schema, API docs, deployment scripts, technical documents?	Yes for the delivered scope: architecture overview, database schema, API notes, deployment scripts and configuration, administrator and user guides, runbooks
7	Repository access from the beginning?	No. The repository is handed over on full payment
8	Credentials, deployment instructions, backup procedures?	Yes, through a signed handover checklist, with credentials rotated at handover
9	Use limited to DSEW; no resale?	Yes. DSEW may not resell, sublicense or share PlotArmour's pre-existing components. We will not resell or disclose DSEW's confidential material

3. Open points for the company and counsel

None of these is decided. No fee is included for any of them.

#	Open point
1	Final legal wording of the ownership and licence clauses above, reviewed by counsel
2	Whether DSEW gets source access to, or modification rights over, PlotArmour's pre-existing components, and on what terms
3	What DSEW can rely on if PlotArmour stops supporting Verity's pre-existing components (continuity or escrow arrangement, if any)
4	Whether general improvements funded by DSEW may be reused by PlotArmour, with DSEW's data and formats excluded
5	Licence extension to affiliates, additional sites or more users, and on what terms
6	Liability, warranty and indemnity clauses; governing law and dispute resolution

This document is not ready for signature until counsel has reviewed it.

Company, Platform and Experience

D & S Electric Works. Who we are, what Verity already is, and what is built for DSEW.

> **Status: DRAFT FOR FINAL REVIEW.** This document states only facts the company has supplied. It makes no claim about clients, references or credentials beyond those listed.

1. Company

Name	PlotArmour
Registered address	31/44 St 3, Block 17, Vishwas Nagar, Delhi, India – 110032
GSTIN	07TQDPS9711L1ZT
Phone	+91 87440 69597
Email	team@plotarmour.in

PlotArmour owns and develops **Verity**, an operations platform for service and manufacturing organisations. Verity is the platform proposed for DSEW.

2. Why Verity suits DSEW, and how much is customised

Verity is a modular platform: the same foundation (data isolation per company, role-based permissions, audit trail, stock ledger, approvals, manufacturing orders, project jobs) carries every module, so new departments, product families and users are added without redeveloping it. DSEW's needs here are engineered, one-off products traced from approved drawing to tested unit, which is what the project configures and extends.

Capability	Exists in Verity today	Built or configured for DSEW in this project
Per-company data isolation, role and permission model, audit trail	Yes	DSEW role set
Sign-in through an identity provider, with an on-premises mode	Yes, tested on our test environments	DSEW configuration
Stock ledger with reversal-only corrections; items, balances, counts	Yes	Used as is
Manufacturing orders from a BOM, route stages, operations, hold and resume, quality checkpoints with evidence, dispatch, reservations	Yes	BOM revisions, measured values, quality gates
Project jobs with milestones, tasks, timesheets, cost lines, quote revisions and approval	Yes	DSEW templates and print formats

Capability	Exists in Verity today	Built or configured for DSEW in this project
Approvals, evidence storage, QR identity for finished units	Yes	DSEW workflows
Install, backup, restore and upgrade scripts, containerised package	Scripts and package exist. A combined database, files and identity restore has **not yet** been rehearsed end to end, and nothing has been tested on DSEW's host	Pre-deployment compatibility check, then rehearsal and acceptance on DSEW's host
Revision-controlled BOM, controlled documents, structured test records	No	Built in both options
Procurement and inventory depth, planning and MRP, traceability, full quality system, finance, HR and payroll, dispatch, contracts, dashboards, documented AI use cases	Partly (stock, purchasing, approvals), not at DSEW depth	Option 2 (Complete ERP); the MVP has the basic procurement, inventory and quality layers only

3. Experience

PlotArmour has completed **10+ past projects, with values from ₹1 lakh to ₹20 lakh**.

We do not claim, in this document, previous deployments specific to ERP, MES, QMS or document-management systems, electrical or project-based manufacturing, Windows Server hosting, barcode-based traceability, or local AI and document intelligence. The platform's manufacturing, quality and document foundations are our own product (section 2). Further detail on past work can be discussed under the confidentiality agreement in section 6.

4. Leadership, teams, duration and support

Name	Role	Based in
Divyom Sharma	Co-Founder & CEO	Delhi
Nakshatra Aggarwal	Co-Founder & CEO	Edinburgh, UK
Shubhankar Rakshit	CFO	Delhi
Ayushmaan Gaur	COO	Chennai

Team	Members
Product Management	Himanshu, Arav and Akshat
Sales & Outreach	Radhika, Kulsoom and Mannat
Marketing	Alvina (Head)

The people and allocation for this project's delivery (project management, technical lead, development, testing and deployment) are confirmed in writing at kickoff.

Estimated duration: about 6 to 8 weeks for the MVP and about 8 to 12 weeks for the Complete ERP, from kickoff (document 2), then the 12-month support period. These are estimates subject to timely DSEW inputs, testing and agreed acceptance criteria.

Support and escalation: during the 12-month period, contact team@plotarmour.in or +91 87440 69597. The service terms (hours, included time and response targets) are in document 1, section 7. **Escalation contact:** Shubhankar Rakshit, CFO.

5. Demonstration

We can demonstrate the existing platform on fictional manufacturing data at your premises or remotely: a project job with quote revisions and approval, a manufacturing order with route stages and quality checkpoints that block completion, a printable traveller, a dispatch pack and a QR identity for a finished unit. We will say clearly which parts exist today and which are built in the project.

6. Confidentiality

We will sign a mutual confidentiality agreement before DSEW shares sensitive documents, and we treat DSEW's requirements and technical information as confidential. DSEW's data is processed on DSEW's own server; any support access is audited and time-limited. At the end of the engagement we delete or return DSEW material we hold.

TECHNICAL SOLUTION DOCUMENT · DOCUMENT 5 OF THE DSEW PROPOSAL PACKAGE

Verity for D & S Electric Works

The complete technical description of the solution: platform specification, installable web application, server footprint and load, requirements, security, recovery, module-by-module design, AI options and the acceptance evidence DSEW receives at go-live.

Prepared for	D & S Electric Works (DSEW), 109, Industrial Area Extension, Rau, Indore 453331, Madhya Pradesh
Prepared by	PlotArmour, 31/44 St 3, Block 17, Vishwas Nagar, Delhi 110032 · GSTIN 07TQDPS9711L1ZT
Contact	team@plotarmour.in · +91 87440 69597
Date	21 October 2026
Companion documents	1 Proposal · 2 Delivery and deployment · 3 Licence and ownership · 4 Company and platform
Classification	Confidential. Prepared for DSEW under the mutual confidentiality agreement

Hardware figures for the Complete ERP with AI search are set in section 7 of this document. Figures stated as targets are confirmed at the pre-deployment survey and proven by the acceptance tests in section 15.

Contents

1 Summary for DSEW management and IT
2 Design principles for a customer-hosted system
3 Solution architecture
4 Verity platform specification
5 Installable web application (PWA) and device reach
6 Server load and performance engineering
7 Minimum and recommended requirements
8 Data architecture and storage growth
9 Security and data protection
10 Availability, backup, recovery and upgrades
11 Module-by-module technical specification
12 Engineering, manufacturing and quality controls
13 Artificial intelligence: options by hardware tier
14 Deployment topology and installation sequence
15 Acceptance tests and go-live gates
16 Requirements traceability matrix
17 Support, continuity and handover
18 Risk register
19 Glossary

HOW TO READ THIS DOCUMENT

Sections 1 to 3 give the shape of the solution. Sections 5 to 7 answer the question DSEW's IT team will ask first: what does this ask of our server, and will it slow anything else down. Sections 4 and 8 to 12 are the platform and module specification. Sections 13 to 17 describe what is optional, how it is proven, and how it is supported.

SECTION 1

Summary for DSEW management and IT

Verity is a modular business operating platform. For DSEW it is configured as an engineering and manufacturing ERP that traces each engineered, one-off product from enquiry through approved drawing, bill of materials, manufacture and testing to dispatch. It runs on one Linux virtual machine on DSEW's own premises, with nothing leaving DSEW's network by default.

1 VM
One Linux virtual machine hosts the whole solution. No cluster, no cloud dependency

4 vCPU · 16 GB
Core ERP sizing for 15 named users. 32 GB when AI semantic search is enabled

0 licences
No paid third-party software. PostgreSQL, Keycloak, Next.js and the rest are open source

Any device
Installable web app (PWA) for office PCs, shop-floor phones and tablets

Thin client
All processing is on the server. User devices only display and capture

19 modules
On one shared foundation in the Complete ERP. 9 in the MVP

What DSEW gets, in technical terms

Application	One web application (TypeScript, Next.js, React) served over HTTPS, installable on any device as a Progressive Web App
Database	PostgreSQL 16, with database-enforced data isolation, a least-privilege application account and an append-only audit trail
Sign-in	Keycloak identity provider on the same VM; password policy, optional multi-factor sign-in, session control
Files	S3-compatible object storage on a separate volume for drawings, certificates, test reports and photographs
Packaging	Containerised with scripted install, backup, restore, upgrade and rollback
Controls	Role-based permissions, approval workflows, revision control, quality gates that block release, immutable audit
Server load	Designed to a stated resource budget, protected by caching, pagination, indexing, background-job scheduling and rate limits, and proven by a load test on DSEW's own VM before go-live (sections 6 and 15)
AI	Optional tiers by hardware, from rule-based detection and full-text search on the same VM up to a separate on-premises GPU machine. AI never approves, releases or changes a record (section 13)
Ownership	DSEW owns its data and can export it at any time. Source code of the DSEW-specific deliverables is handed over on full payment (document 3)

THE SERVER-LOAD ANSWER IN ONE PARAGRAPH

Verity is not a heavy system. It is one application process and one database, sized for a four-core, 16 GB VM and not a data centre. User devices carry no processing load. Report generation, file indexing and any AI indexing run as scheduled background work outside office hours, with limits so that a heavy job can never starve interactive users. DSEW's other workloads on the same Hyper-V host are protected by fixed CPU and memory allocation to the VM. A load test against agreed response-time targets on DSEW's actual VM is a go-live gate.

SECTION 2

Design principles for a customer-hosted system

Most business software is built to run in a vendor's cloud, where capacity can be added at will. A system that runs on the customer's own server has the opposite constraint: the hardware is fixed, it is shared with other work, and the customer's IT team carries the operational risk. Verity's deployment design for DSEW follows eight principles.

#	Principle	What it means for DSEW's server
1	Small, known footprint	A fixed set of five services on one VM, each with a memory budget (section 6). There is no message-broker cluster, no search cluster and no second database engine to feed.
2	Server does the work, device displays	Office PCs, phones and tablets render pages and capture input. Nothing computational is pushed to them, so old or low-end devices work.
3	Interactive work first	Anything slow (reports, exports, indexing, backups, AI) runs as scheduled background work with concurrency limits and runs outside shift hours by default.
4	Bounded queries	Every list is paginated and indexed. No screen loads an unbounded table. Search is database-indexed.
5	Fail closed	If tenant context or permission is missing, queries return nothing and writes are rejected. A fault never widens access.
6	Everything can be put back	Scripted backup, tested restore, versioned upgrades with rollback. Recovery is rehearsed before go-live, not merely described.
7	Data stays on the premises	No cloud service is required to run, and none receives DSEW data by default. Outbound access is needed only for install and update, and can be replaced by offline media.
8	Standardise the foundation, configure the behaviour	Identity, permissions, audit, numbering, workflow and storage are one shared foundation. DSEW's specific behaviour is configuration and modules on top, so later additions do not require redevelopment.

SECTION 3

Solution architecture

Verity is built as layers. Each layer has one job, and an upper layer cannot bypass the one below it. This is what allows new capabilities to be added later without redesigning the foundation.

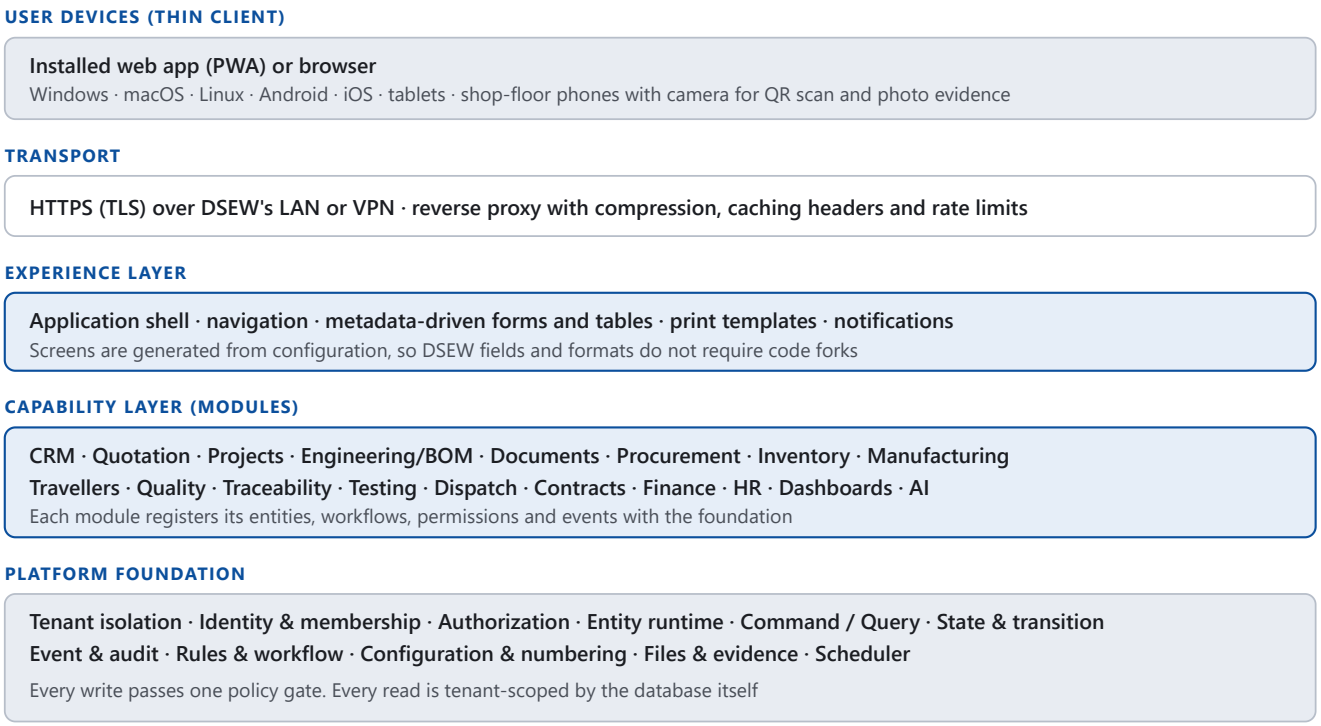


Figure 1. Layers of the solution. Modules are added in the capability layer without changing the foundation.

3.1 Runtime components

Component	Technology	Role
Application	TypeScript, Next.js, React	Serves the interface and the permissioned API. Runs the business logic: commands, queries, state transitions, rules and events
Database	PostgreSQL 16	System of record. Row-level security enforces company isolation. Holds records, audit trail, the event log and full-text indexes
Identity	Keycloak (OpenID Connect)	Sign-in, password policy, optional multi-factor sign-in, session lifetime. Application roles remain in Verity
File storage	S3-compatible object store	Drawings, certificates, reports, photographs. Held on a separate volume so file growth never competes with the database
Reverse proxy	Standard TLS-terminating proxy	HTTPS, compression, static-asset caching, rate limiting, security headers
Packaging	Docker and Compose	Reproducible install; versioned images; scripted upgrade and rollback
Scheduler	Built in, one trigger	Runs background work (reminders, escalations, reports, indexing, backups) on a schedule with concurrency limits

3.2 Path of a single request

1. The user's device sends an HTTPS request carrying a signed session issued after Keycloak sign-in.
2. The application derives the **company (tenant) context from the authenticated session**, never from anything the browser sends in the request body.
3. For a change, the request becomes a **command**. One policy gate checks the user's permission (action, entity, scope) and the record's current state.
4. Inside one database transaction the tenant is set, the state change is written, the **audit entry and event** are recorded, and the transaction commits or rolls back as a whole.
5. For a read, a **query** returns only the rows that tenant and scope permit; the database applies the same filter even if the application code were wrong.
6. The response returns a small payload. Large lists are paged; files are streamed from the object store.

3.3 Extension model

A new capability is registered, not patched in. It declares its entities, workflows, permissions, events, screens and the other capabilities it depends on. DSEW's later additions (a new product family, an inspection format, a new department) are configuration or added modules; the foundation, the workflow engine and the authorisation model are not modified, and no code is forked per client.

SECTION 4

Verity platform specification

This section defines the platform foundation on which all 19 modules rest. These are the building blocks that make the system consistent, auditable and extensible.

4.1 Identity and data model

Concept	Definition
Tenant	The root data-isolation boundary. DSEW is one tenant. Every record belongs to exactly one tenant and no record references another tenant's data
Organization	A business unit inside the tenant (for example a plant or a department). Permissions can be scoped to it
Party	One record per person or business: customer, supplier, employee, contact. A single identity, never duplicated across modules
User	A person who signs in. One-to-one with a Party
Membership	A user's place in an organization, which carries role assignments
Location	A site, store or bay. Distinct from an address and from a geofence
Resource	A schedulable unit backed by a person or an asset: a machine, a tester, a technician
Asset	Physical equipment: machines, test instruments, tools
Work	A unit of work to be executed: a manufacturing job, an inspection, a project task
Evidence	Immutable field data: photographs, measured values, signatures, scanned documents
Contract	An agreement with its terms, obligations and exposures

4.2 Authorization

- **Permissions are Verb + Entity + Scope.** For example *approve · Quotation · Organization* or *release · BillOfMaterials · Tenant*. There are no hard-coded roles such as "manager" in the code; roles are named sets of permissions that DSEW defines.
- **Scopes** narrow where a permission applies: tenant-wide, one organization, or one location. Additional axes such as team or resource are handled by configuration.
- **One decision point.** Every command, from the screen, the API, a scheduled job or an AI assistant, passes the same policy check. There is no back door and no elevated service path.
- **Separation of duties** is configurable: DSEW chooses who drafts, reviews, approves and releases, and whether self-approval is ever allowed.
- **The AI channel acts as the calling user.** Whatever an AI feature does, it does with that user's own permissions and nothing more (section 13).

4.3 State and transition

Business records move through configurable **state machines**. Every state belongs to one of six behavioural categories, so reports and timers can read behaviour without depending on DSEW's own labels.

Category	Meaning	Example DSEW states
Draft	Being prepared; not yet in force	Quotation draft, BOM draft
Pending	Waiting for a decision or input	Awaiting approval, awaiting material
Active	In progress	In winding, in assembly, in testing
Blocked	Cannot proceed until a condition clears	On hold for NCR, awaiting drawing revision
Completed	Execution finished (terminal)	Tested and passed
Cancelled	Abandoned (terminal)	Enquiry lost, order cancelled

Completed and Closed are different. Completed means the work is done. Closed is an administrative step that makes the record **permanently read-only**. A closed record cannot be edited; later rework creates a linked child record. This is how "what was approved is what was built" is enforced by the system and not by habit.

4.4 Commands, queries, events and audit

Command	A request to do something: create a quotation, release a BOM, issue material. Validated, authorised, state-checked and applied in one transaction
Query	A read of current state, always tenant-scoped and permission-scoped
Event	A fact that something happened (*BOM released*, *inspection failed*). Other modules and notifications react to events, so modules stay loosely coupled
Projection	A derived read view (dashboards, ageing, stock balance) built from events and records
Audit trail	For every change: who, what, when, from which channel, previous and new values. Written in the same transaction as the change; the application cannot alter or delete it
Failed attempts	Refused commands are recorded as metadata, so repeated denied attempts are visible to the administrator

4.5 Rules, workflow and configuration

- **Workflow definitions** are data: nodes (steps), edges (transitions) and conditions (for example "approval required above a value", "all mandatory checks passed").
- **Custom fields** add DSEW-specific attributes (kVA rating, impedance, cooling class, customer drawing number) to any record without schema forks.
- **Document numbering** is configurable per document type with prefix, year, site and sequence, allocated safely under concurrent use so numbers are never duplicated.
- **Forms and tables** are generated from metadata, so adding a field or a column is configuration with permission, validation and audit already applied.
- **Print templates** produce DSEW's branded quotation, traveller, challan, packing list, certificate and report formats.

- **Capability activation**: modules are switched on per tenant, and version pins mean DSEW upgrades on its own schedule.

4.6 Application programming interface

A permissioned REST API exposes the same commands and queries the screens use, under the same policy gate. DSEW can read its data programmatically for reporting. No external system integration is included in scope (document 1, section 4).

SECTION 5

Installable web application (PWA) and device reach

Verity is delivered as a Progressive Web App. It opens in any modern browser, and on any device it can also be installed with one tap so it behaves like a native application, with no app store, no per-device installer and no separate mobile build to maintain.

5.1 Why this matters for DSEW

One build, every device

The same application serves office PCs, tablets and shop-floor phones. There is no Android build, iOS build and Windows build to keep in step, so every user sees the same version on the day it is upgraded.

Zero client installation work

Users open an internal address and choose "Install". IT does not push software to devices. Removing it is just as simple, and a lost phone holds nothing once its session is revoked.

Light on the device

Pages are server-rendered and compressed, so a modest phone or an older office PC performs well. The installed app is a shortcut to the same web application, not a second copy of the system.

Light on the server

The application shell, fonts, icons and scripts are cached on each device after the first visit. Later visits fetch only data, which cuts repeated traffic and CPU work on DSEW's server.

5.2 What is delivered

Capability	Specification
Web app manifest	Name, icons for all sizes, theme colour and standalone display, so the installed app opens in its own window without browser chrome
Service worker	Caches the application shell and static assets. Business data is never served stale from cache for a transaction; it is always read from the server
Secure by default	PWA features require HTTPS. The internal TLS certificate arranged in the pre-deployment check (document 2) satisfies this
Camera capture	Photograph upload for inspection evidence and QR scanning with the device camera. Keyboard-style hardware scanners also work as ordinary input
Responsive layouts	Phone, tablet and desktop layouts. Touch targets are sized for shop-floor use. Tables adapt to narrow screens
Update behaviour	When DSEW upgrades the server, the next load picks up the new version and the user is prompted to refresh. No device-by-device update
Connection state	If the network drops, the application shows a clear "reconnecting" state and protects unsaved entries from being lost. It does not silently queue business transactions that could conflict
Accessibility	Keyboard operation, visible focus, screen-reader labels, and text contrast checked to WCAG AA in light and dark appearance

5.3 Supported devices

Device	Browsers	Install method	Typical DSEW use
Windows PC / laptop	Chrome, Edge, Firefox	Install icon in the address bar (Chrome, Edge)	Office, engineering, stores, accounts
macOS / Linux	Chrome, Edge, Firefox, Safari (macOS)	Install from the browser menu	Engineering, management
Android phone / tablet	Chrome, Edge	"Install app" prompt	Shop floor, inspection, stores, QR scan
iPhone / iPad	Safari	Share → Add to Home Screen	Management, shop floor, inspection
Thin client / kiosk	Chrome or Edge, full-screen	Shortcut to the internal address	Shared terminal at a workstation

5.4 Network use on the premises

In-LAN use needs no internet after installation. Shop-floor phones reach the server over the premises Wi-Fi to the internal HTTPS address. Remote use goes only through DSEW's VPN. Page weight after the first load is small, because the shell is cached and responses are compressed, so ordinary Wi-Fi is sufficient.

SECTION 6

Server load and performance engineering

DSEW will host Verity on its own server, so the load Verity places on that server is a design requirement, not an afterthought. This section sets out where load comes from, how it is held down, how much of the VM each part may use, and how it is proven before go-live.

6.1 Where load comes from, and how each source is controlled

Source of load	Why it can be heavy	Control built into the solution
Interactive screens	Large lists and joins across modules	Server-side pagination, indexed queries, bounded page size, selective columns. No screen loads an unbounded table
Search	Scanning free text across records and documents	PostgreSQL full-text indexes maintained on write; search never scans tables
Reports and dashboards	Aggregations over many records	Precomputed projections refreshed in the background; heavy reports queued to run off-hours; result caching
File upload and download	Large drawings and scans	Streamed to and from the object store, not buffered in memory; size limits per type; separate volume
Background jobs	Reminders, escalations, indexing, exports	One scheduler, concurrency caps, run windows outside shift hours, jobs yield to interactive traffic
Concurrency spikes	Many users at shift start or approval deadlines	Database connection pooling in transaction mode; request rate limits; short transactions
Backups	Disk and CPU during dump	Scheduled outside working hours, low priority, written to the separate backup target
AI indexing (optional)	Embedding generation is CPU-intensive	Background only, throttled, paused when interactive load is high (section 13)

6.2 Resource budget for the core VM (4 vCPU, 16 GB)

The VM is given fixed resources so it cannot grow into the memory or CPU that DSEW's other workloads on the Hyper-V host need, and the budget below leaves headroom inside the VM so that peaks do not push it into swapping.

Service	Planned memory	Notes
Operating system and tools	≈ 1.0 GB	Ubuntu Server LTS, no desktop environment
Application (Node runtime)	≈ 1.5 to 2.5 GB	Worker count tuned to the cores; bounded heap
PostgreSQL 16	≈ 3.0 to 4.0 GB	Shared buffers plus working memory, bounded per connection
Keycloak	≈ 1.0 to 1.5 GB	Authentication only; light for 15 users
Object store and proxy	≈ 0.5 to 1.0 GB	Stream-oriented

Service	Planned memory	Notes
Planned total in use	**≈ 7 to 10 GB**	Remaining memory serves as the operating system's file cache, which speeds up database reads

PLANNING FIGURES

The values above are the design budget used to size the VM. The actual footprint is measured on DSEW's VM during installation and recorded in the acceptance report (section 15).

6.3 Workload model used for sizing

Parameter	Planning value
Named users	Up to 15 (scales with the sizing table in 6.4)
Concurrent active users	Up to 10 to 12 at peak, since not all named users are active together
Typical interaction	One page action every 20 to 40 seconds per active user; most are reads
Peak events	Shift start (many sign-ins and dashboard views); month-end (reports); approval cut-offs
Document volume	Drawings, certificates, reports and photographs attached to projects and jobs (section 8)

6.4 Sizing by users and data

Scenario	vCPU	Memory	Disk	Notes
Up to 15 users, core ERP	4	16 GB	Within the 500 GB SSD	The offered configuration
Up to 15 users, with AI semantic search	4 to 6	32 GB	Within the 500 GB SSD, plus index storage	Embedding model and vector index compete for memory
Up to 30 users	6	24 to 32 GB	As above	Quoted separately; no redesign needed
Up to 60 users	8	32 GB	Larger SSD	Quoted separately; same architecture
Interactive AI drafting	n/a	n/a	n/a	Separate on-premises GPU machine on the LAN (section 13); the ERP VM is not resized for it

6.5 Response-time targets proven at acceptance

These are the targets the load test is run against on DSEW's VM at 15 concurrent sessions with realistic mixed actions. They are agreed with DSEW at requirements sign-off and the results are recorded in the acceptance report.

Action	Target (95th percentile)
Open a list page of up to 50 rows (filtered, sorted)	under 1.5 s
Open a record with its linked data	under 1.5 s
Save a record or transition a state	under 1.0 s
Full-text search across records and documents	under 2.0 s

Action	Target (95th percentile)
Management dashboard load	under 3.0 s
Upload of a 10 MB file over the LAN	under 5 s
Sustained VM CPU during the test	average under 60 %
Memory in use, no swapping	within the planned budget

6.6 Protecting DSEW's other workloads

- **Fixed allocation:** the VM is created with static memory and a defined number of virtual processors, so Verity cannot take more of the host than agreed.
- **Disk separation:** database, files and backups are on separate volumes, so file growth or a backup cannot starve database input and output.
- **Quiet background:** backups, indexing and large reports run in a scheduled window that DSEW sets, at low priority.
- **Visibility:** a health view for the administrator shows CPU, memory, disk, database size, queue depth and slow queries, with alert thresholds, so capacity is a managed number and never a surprise.
- **Growth path:** if DSEW outgrows the VM, resources are increased on the same architecture. There is no re-platforming.

SECTION 7

Minimum and recommended requirements

Final figures are confirmed in the pre-deployment survey (document 2, section 6). Where this section differs from the hardware line in document 2 for the Complete ERP with AI search, this section governs.

7.1 Server (one virtual machine)

Item	Minimum (core ERP, 15 users)	Recommended	Complete ERP with AI search
Virtual processors	4	4 to 6	4 to 6
Memory	16 GB, static	16 GB	**32 GB**, static
System and database disk	500 GB SSD	500 GB mirrored SSD	500 GB mirrored SSD plus index space
Document volume	Separate volume, sized with DSEW	Separate volume	Separate volume
Operating system	Ubuntu Server LTS (Linux guest). Not run natively on Windows Server and not run through Docker Desktop or WSL		
Host	Hyper-V role enabled with hardware virtualisation on, or a small dedicated host		

7.2 Network and security services

Item	Requirement
Address and name	A fixed IP address and an internal DNS name for the VM
LAN	Gigabit LAN between the server and office switches; reliable Wi-Fi coverage on the shop floor
TLS certificate	A certificate for the internal name, trusted by office PCs and phones (public CA or DSEW's internal CA). Required for sign-in security and for the installable web app
Remote access	DSEW's VPN, only if remote use is wanted
Internet	Controlled outbound access for installation and updates, or offline media. None at runtime
Time	Time synchronisation (NTP) so audit timestamps are correct

7.3 Backup and power

Item	Requirement
Backup target	A separate device or share, at least equal to the data size, ideally with a second off-site or offline copy
Power	UPS sized for a clean shutdown of the host and the VM

7.4 User devices

Item	Requirement
Office PC	Any PC from the last several years with a current Chrome, Edge or Firefox. No local software installation required
Phone / tablet	Android 9 or later with Chrome, or iOS 16 or later with Safari; rear camera for photographs and QR scan
Screen	1366 × 768 or larger recommended on desktop; any phone or tablet size is supported

7.5 People

A named DSEW product owner (about 3 to 4 hours a week), leads for quality, engineering, stores and finance, and a named IT contact with the rights needed during the survey and installation (document 2, section 12).

7.6 Optional AI hardware

Only if DSEW chooses interactive AI drafting: a separate small workstation with a 16 GB GPU on the LAN. It is not part of the VM and not included in the fee. Details and tiers are in section 13.

SECTION 8

Data architecture and storage growth

8.1 Data domains

Domain	Main records
Identity and access	Tenant, Organization, Party, User, Membership, Role, Permission, Grant
Commercial	Customer, Contact, Enquiry, RFQ, Quotation and revisions, Sales Order, Contract, Retention, Liquidated-damages terms
Project	Project, Milestone, Task, Budget, Risk, Change request
Engineering	Product, Component, BOM and revision, Engineering Change Request and Order, Drawing
Documents	Controlled document, Revision, Transmittal, Acknowledgement, Numbering series
Supply chain	Supplier, Purchase Request, RFQ, Purchase Order, Goods Receipt, Stock ledger, Reservation, Warehouse and bin
Manufacturing	Manufacturing job, Operation, Route, Traveller, Work centre, Labour, WIP, Subcontract
Quality and test	Inspection plan, Checklist, Result, NCR, CAPA, Calibration, QAP/ITP, Test record, Certificate
Traceability	Batch, Heat, Serial, QR identity, Genealogy link
Logistics	Dispatch plan, Packing list, Challan, Shipment
Finance and HR	Invoice, Payment, Receivable, Payable, Profitability; Employee, Attendance, Leave, Payroll run, Payslip
Platform	Audit trail, Event log, Notification, Scheduled job, Configuration

8.2 Where storage growth actually comes from

Structured records are small. Storage growth is dominated by **files** (drawings, scans, photographs, test reports), which is why files live on their own volume and not inside the database.

Store	Growth driver	Planning basis
Database (records, audit, events, search indexes)	Transactions and revisions	Tens of thousands of records a year at DSEW's volume; a few gigabytes over five years including indexes and audit
Document volume	Files per project and job	Sized with DSEW from its sample documents; typically the largest consumer
Vector index (AI semantic search only)	Chunks of indexed documents	Proportional to indexed text, not to image or CAD file size
Backups	Retention policy	At least equal to data size on the backup target, plus retention copies

PLANNING FIGURES

Growth values are planning assumptions from typical engineering-order volumes. The document-volume size is finalised in the survey from DSEW's real samples, and a capacity report is part of handover.

8.3 Data lifecycle and export

- **Retention:** closed records stay readable and locked; retention periods for documents are configured per type.
- **Export:** a documented export of all records as CSV and all files in original form with an index, at handover and once on request during the 12-month period.
- **Data location:** all data is held on DSEW's server. Support access is audited and time-limited.
- **Deletion:** at the end of an engagement, PlotArmour deletes or returns any DSEW material it holds.

SECTION 9

Security and data protection

Security is part of the foundation. It is applied to every operation from the first release, not added as a layer afterwards.

9.1 Controls by layer

Layer	Control
Transport	TLS for all traffic; HTTP redirected to HTTPS; modern protocol versions only; security headers (content security policy, frame protection, referrer policy)
Authentication	Keycloak with password policy and lockout; optional multi-factor sign-in, enabled on request; configurable session lifetime and idle timeout; immediate session revocation
Authorization	Verb + Entity + Scope permissions checked on the server for every command; no reliance on hidden screens; one policy gate for all channels
Tenant isolation	Company context comes from the authenticated session, never from the request. Row-level security in PostgreSQL enforces isolation even if application code were wrong; with no scope set, queries return nothing and writes are rejected
Database account	The application connects as a least-privilege account that cannot bypass row-level security. Start-up refuses a privileged account. A separate account is used only for schema migration
Audit	Append-only trail of every change with actor, time, channel and before/after values; cannot be edited or deleted by the application
Input handling	Server-side validation; parameterised queries; output encoding; protection against cross-site request forgery; upload type and size checks
Secrets	Held in a protected environment file (restricted permissions, owned by a named administrator), never in the database or in source. A sealed offline copy is held by DSEW's owner
Files	Access through the application's permission check; files are not publicly addressable
Backups	Encrypted at rest on the backup target; restore tested
Disk	Full-disk encryption recommended on the VM and backup target, confirmed at acceptance
Network exposure	Only the HTTPS port is exposed. Database, identity and object store are bound to the VM's internal network and not reachable from the LAN
Software supply	Pinned, versioned releases; dependency vulnerability review at each release; tested upgrade and rollback
Support access	Remote, time-limited, audited, and only with DSEW's permission

9.2 Rule for every change to data

For each state-changing operation the design records: who is acting, the permission required, the scope, the preconditions, the state change, the audit entry, the event emitted, and the behaviour on failure. A change that cannot answer all eight is not released.

9.3 AI and data protection

Nothing leaves DSEW's server to an external AI service unless DSEW explicitly enables that service. Local AI features run inside the signed-in user's own permissions, so a user can never retrieve through AI what they could not open directly. AI is never an approver and never changes a record on its own (section 13).

9.4 Scope boundary

Independent penetration testing and statutory certification are outside this scope. They can be arranged separately.

SECTION 10

Availability, backup, recovery and upgrades

10.1 Backup design

Component	Method	Frequency (default, agreed with DSEW)
Database	Consistent logical dump; optional continuous log archiving for finer recovery	Nightly full; optional archiving for a shorter data-loss window
Files	Incremental copy of the document volume	Nightly
Identity configuration	Realm export	Nightly and before every change
Application configuration	Environment and version record	With every upgrade
Retention	Daily, weekly and monthly generations	Set with DSEW

10.2 Recovery objectives (proposed, agreed with DSEW)

Objective	Default target	Tighter option
Data-loss window (RPO)	Up to 24 hours (nightly backup)	Up to 15 minutes with log archiving
Time to restore (RTO)	Within 4 hours on the same or a replacement VM	Agreed after the restore rehearsal

A **combined restore** of database, files and identity configuration onto a clean environment is rehearsed during the project (after Phase 2) and again as a final test before go-live. The restore time measured in the rehearsal replaces the default target in the handover documents.

10.3 Failure behaviour

Failure	Behaviour
Identity service unavailable	Existing sessions continue; new sign-ins fail with a clear message; quality gates remain enforced
Object store unavailable	Records remain usable; new uploads fail clearly; no record claims a file that was not stored
Application restart	Short interruption; no partial writes because every change is one transaction
Power loss	Database recovers to the last committed transaction on restart; UPS provides clean shutdown
Disk filling	Alerts at configured thresholds before the limit; document growth is on a separate volume
Network interruption on a device	The application shows connection state and protects unsaved input

10.4 Upgrades and rollback

1. Release notes and the changes to DSEW's installation are shared in advance.
2. A backup is taken automatically before the upgrade.
3. The upgrade is applied with database migrations in a defined order.
4. Health checks confirm the new version. If they fail, the scripted **rollback** restores the previous version and data state.
5. Upgrades are applied at a time DSEW chooses, typically outside working hours.

SECTION 11

Module-by-module technical specification

All modules share the foundation in section 4, so each inherits permissions, audit, numbering, search and event notifications. The table lists the main records, workflow states, controls and links to other modules for the Complete ERP. The MVP includes the basic layers described in document 1, sections 3 and 5.

#	Module	Main records and workflow	Controls and rules	Links to
1	CRM, enquiry, quotation, sales order	Customer, contact, enquiry, RFQ, tender, quotation with revisions, sales order. Draft → Pending approval → Active → Won / Lost	Revision history; approval above configured value; alternative offers; PO comparison; retention and LD terms captured; follow-up reminders	Project, BOM, finance, documents
2	Advanced project management	Project, work breakdown, milestone, task, budget, risk, change. Draft → Active → Blocked → Completed → Closed	Dependencies; budget versus actual; change control; status roll-up	Orders, manufacturing, finance, contracts
3	Engineering and product lifecycle	Product, component, multi-level BOM and revision, ECR and ECO. Draft → In review → Released → Superseded	A released revision cannot be edited; every order records its revision; changes follow ECR/ECO approval	Manufacturing, procurement, travellers
4	Document management and controlled workflows	Controlled document, revision, transmittal, acknowledgement. Draft → Review → Approved → Issued → Obsolete	Configurable numbering; drafter, reviewer, approver and releaser roles; full revision and distribution history; search by number, title, content	All modules
5	Manufacturing execution	Manufacturing job, operation, route stage, work centre. Planned → In progress → On hold → Completed	Job identifies its BOM; material availability shown; operations tracked; hold and resume with reason; WIP; subcontract tracking	BOM, inventory, traveller, quality
6	Travellers and process documentation	Traveller generated from job and route; checkpoints and sign-offs	BOM-driven content; QR code on printout; sign-off by role; customer-facing formats	Manufacturing, quality, traceability
7	Quality management system	Inspection plan, checklist, result, NCR, CAPA, calibration, audit, QAP/ITP	Mandatory checks block release; NCR closes only after CAPA verification; evidence attached to results	Manufacturing, dispatch, testing, suppliers
8	Barcode, QR and traceability	Batch, heat, serial, QR identity, genealogy	Unit-to-component genealogy; scan to open the record; QR verification page for a finished unit	Inventory, manufacturing, quality, dispatch
9	Procurement and vendor management	Supplier, purchase request, RFQ, comparative statement, PO, receipt. Draft → Approved → Ordered → Received	Approval workflow by value; supplier performance; subcontract orders; match to receipt	Inventory, finance, quality
10	Inventory and warehouse	Item, stock ledger, reservation, warehouse and bin, count	Ledger with reversal-only corrections (no silent edits); valuation; reconciliation; ageing; batch/heat/serial	Procurement, manufacturing, finance

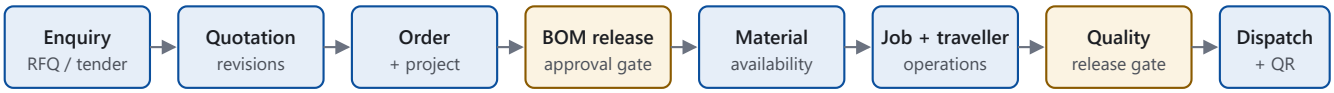
#	Module	Main records and workflow	Controls and rules	Links to
11	Production planning and basic MRP	Demand, supply, planned order, shortage	BOM-based requirements; shortage recommendations; capacity view; scheduling by work centre	BOM, inventory, procurement, manufacturing
12	Testing and equipment	Test template, test record, certificate, instrument, calibration, maintenance	Tests per product family with acceptance limits; automated report; failed test blocks dispatch; calibration-due alerts	Quality, dispatch, assets
13	Finance and accounting	Invoice, payment, receivable, payable, GST rates, profitability	Invoices from orders and milestones; payments reduce balances; receivables reconcile; job profitability reconciles to costs	Sales, procurement, projects, payroll
14	HR and payroll	Employee, attendance, leave, shift, timesheet, salary structure, payroll run, payslip	Payroll from DSEW-supplied statutory components; a sample month reproduces DSEW's own calculation; competency tracking	Projects, manufacturing labour
15	Dispatch and logistics	Dispatch plan, packing list, challan, shipment. Ready → Packed → Dispatched	Dispatch released only after quality approval and test pass; partial deliveries; shipment tracking	Quality, inventory, finance
16	Contracts, compliance and risk	Contract, obligation, bank guarantee, LD exposure, insurance, NDA, risk register	Obligation due-date alerts; LD exposure tracking; expiry reminders	Orders, projects, finance
17	Dashboards and business intelligence	Role-based dashboards, KPI views	Precomputed projections; profitability, ageing, cash flow, supplier and resource analytics; respects permissions	All modules
18	Notifications and workflow automation	Rules, triggers, escalations, templates	Event-driven; in-app and e-mail through DSEW's mail server; escalation chains; automated documents	All modules
19	AI (documented use cases)	Search, retrieval, classification, gap detection, summaries, drafts	Inside the caller's permissions; never an approver; acceptance threshold per use case on agreed samples	Documents, quotation, quality

SECTION 12

Engineering, manufacturing and quality controls

DSEW builds engineered, one-off products. The central requirement is that what was approved is what is built, and nothing leaves the factory without its checks. Verity enforces this through five connected controls.

12.1 From enquiry to dispatch: the controlled chain



Amber steps are gates: they block the next step until the condition is met

Figure 2. Linked records from enquiry to dispatch, with two blocking gates.

12.2 The five controls

Control	How it works
Revision lock	A released drawing or BOM revision cannot be edited. A change creates a new revision through the approval path. A job records the revision it was built to.
Approval separation	Drafting, review, approval and release are different permissions. DSEW decides whether one person may hold more than one.
Quality gate	A failed or missing mandatory inspection or test blocks quality release. An NCR closes only after CAPA verification. Dispatch is blocked until release.
Immutable evidence	Measured values, photographs and signatures are stored as evidence that cannot be altered, with the user, time and device.
Closed means closed	When a job or order is closed, it is permanently read-only. Rework is a new linked record, so history is never rewritten.

12.3 Traceability

Each finished unit carries a QR identity. Scanning it opens the unit's record: the sales order, BOM revision, components with batch or heat numbers, route and operations, inspection and test results, and dispatch details. A forward trace answers "which units used this batch"; a backward trace answers "what went into this unit". An optional public verification page shows a controlled subset for the customer.

12.4 Stock integrity

Inventory is a ledger: every receipt, issue, transfer and adjustment is a posted entry. Corrections are made by reversal with a reason, never by silently changing a balance, so the quantity on hand can always be explained from its history.

SECTION 13

Artificial intelligence: options by hardware tier

DSEW asked that confidential client data must not leave its server. The AI design therefore runs on DSEW's premises, with capability that scales with the hardware DSEW chooses. DSEW can start with no extra hardware and add more after seeing real search and rules working on its own documents.

13.1 Tiers

Tier	What it is	Where it runs	What it delivers
Tier 0	No model. Search and rules	The ERP VM; no extra hardware	Full-text search across documents and records; rule-based detection of missing documents and incomplete records; template and pattern extraction of drawing number, revision and PO number. Finds past quotations, POs, BOMs and jobs quickly and flags gaps
Tier 1	Small embedding model with a vector index	The ERP VM at 32 GB memory	Semantic search, for example "find the 100 kVA cast resin jobs", across the documents DSEW is permitted to see. Indexing runs in the background
Tier 2	Small language model on CPU	The ERP VM, background only	Overnight management summaries and batch tagging. Not offered for interactive drafting, where response time would be too slow
Tier 3	Local language model on a GPU machine	A separate small workstation with a 16 GB GPU on DSEW's LAN	Interactive drafting of quotations, work instructions, inspection plans and reports from approved templates, at usable speed. Verity calls it over the LAN, so data stays on the premises
Tier 4	External AI service	Off DSEW's server	Higher quality for non-confidential text only. Off by default; enabled only by DSEW's explicit decision; usage is charged by the provider

13.2 What is included and what is optional

- **Included with the Complete ERP (module 19):** Tier 0 and Tier 1, the integration of AI into the platform, and the documented use cases with their acceptance thresholds.
- **Optional, chosen by DSEW:** Tier 2 background summaries; Tier 3 GPU machine for interactive drafting; Tier 4 external service.
- **Not in the fee:** GPU hardware and any external service charges. Tier 3 hardware is priced after the use-case survey, so DSEW chooses it with real results in hand.
- The GPU machine is a **separate box**. Sharing a GPU with the Linux VM on the Windows host is not recommended, because it is fragile and complicates rollback.

13.3 Guardrails that apply at every tier

Guardrail	Specification
AI is never an approver	It does not approve quotations, release documents, alter records, authorise quality releases or act autonomously. No essential workflow depends on it

Guardrail	Specification
Inside the user's permissions	An AI feature acts as the calling user, with the same tenant, role and permissions, passing the same policy gate. It can never retrieve what the user could not open directly
Provenance	Results show the source records or documents they came from. Drafts are labelled and need a human to save and submit them
Acceptance thresholds	Each documented use case has defined inputs, outputs and an agreed threshold on DSEW's sample documents; it is accepted only if it meets the threshold
Data stays local	No document or record is sent off the server unless DSEW enables Tier 4
No training on DSEW data	Custom model training is excluded. DSEW's data is used only for retrieval at the moment of the request

13.4 Load control for AI

Indexing and any CPU-based model work run as background jobs with a concurrency limit and a daily window. They pause when interactive load rises, so AI can never slow the ERP for people working. Tier 3 inference runs on a different machine and adds no load to the ERP VM.

> **GO-LIVE GATE FOR AI**
>
> The selected AI tier must run within the latency agreed with DSEW, measured on the surveyed hardware, and each documented use case must meet its acceptance threshold on the agreed sample documents.

SECTION 14

Deployment topology and installation sequence

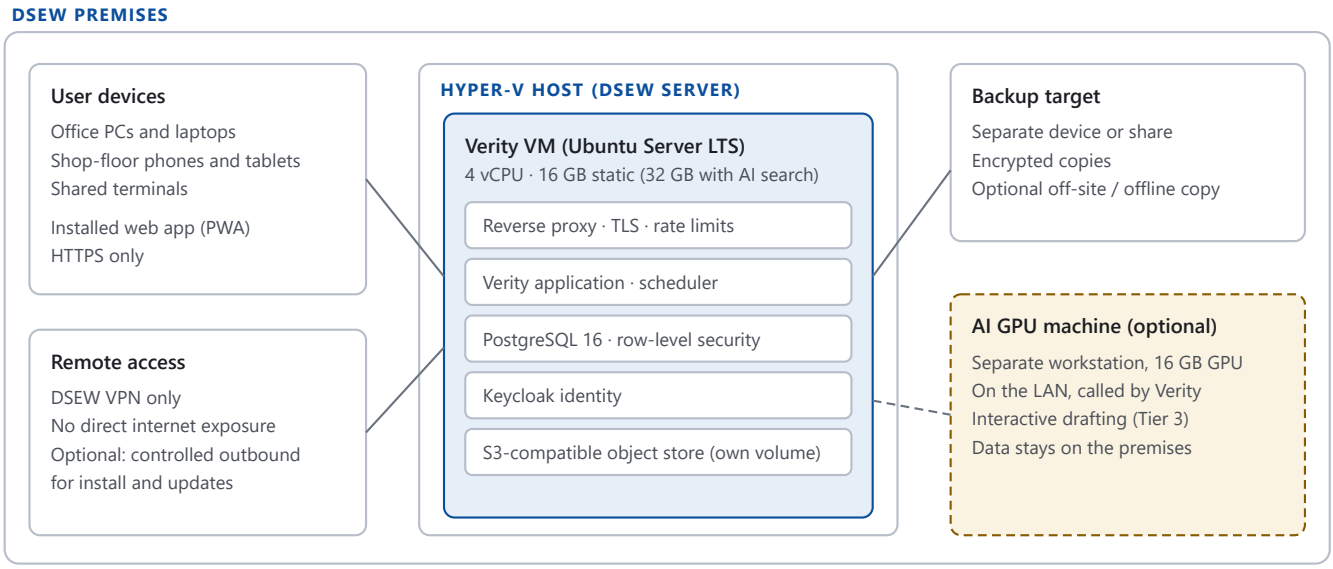


Figure 3. Deployment on DSEW's premises. The dashed AI machine is optional.

14.1 Installation sequence

#	Step	Outcome
1	Pre-deployment compatibility check	Non-intrusive review of the host, virtualisation, network, certificate, backup target and policies. Pass or fail recorded per dependency; nothing on the host is changed
2	VM creation	Ubuntu Server LTS VM with the agreed static resources and volumes
3	Platform installation	Scripted install of the containers, database, identity provider and object store; secrets generated and sealed
4	HTTPS and DNS	Internal name and trusted certificate; PWA install verified from a real phone and a PC
5	Configuration	DSEW roles, numbering, workflows, custom fields, print templates and master-data import
6	Backup schedule	Jobs scheduled; first backup and restore rehearsal
7	Monitoring	Health view and alert thresholds set
8	Load and acceptance tests	Performance test against section 6.5 and UAT scripts, results recorded

14.2 Environments

A company-hosted test environment serves demonstrations and early hands-on use. The DSEW VM is installed at kickoff, used for UAT, and becomes production only after the gates in section 15 pass. A second production VM is not included.

SECTION 15

Acceptance tests and go-live gates

Every claim in this document is backed by a test that DSEW sees run and signs off. Production use begins only when all the gates pass, and the 12-month support period starts on the date they pass and the handover is signed.

15.1 Go-live gates

#	Gate	Evidence DSEW receives
1	Server survey and compatibility	Written pass/fail record per dependency
2	Installation and HTTPS on DSEW's host	Working access from office PCs and a real-phone upload; installed web app confirmed on supported devices
3	**Performance and resource acceptance**	Load test on DSEW's VM at 15 concurrent sessions against section 6.5; measured CPU, memory, disk and response times; resource footprint recorded
4	Combined backup and restore	Restore of database, files and identity configuration onto a clean environment; measured restore time
5	Outage behaviour	Identity and object-store outage tests: sessions continue, new sign-ins or uploads fail clearly, quality gates stay enforced
6	UAT scripts	Signed scenario results with date, tester and outcome
7	Data export reconciliation	Export reconciles with the system
8	Security checks	Tenant-isolation test, permission test, audit-trail immutability test, TLS configuration check
9	**AI latency and threshold** (if an AI tier is selected)	Selected tier runs within the agreed latency on the surveyed hardware; each documented use case meets its threshold on agreed samples

15.2 UAT scenarios

1. Enquiry to quotation to order to project.
2. BOM draft, review, approval and release; a released revision cannot be edited.
3. Material availability, purchase, receipt and issue to a job.
4. Manufacturing job with traveller; operations recorded; hold and resume.
5. Failed mandatory test blocks quality release and dispatch.
6. Dispatch released after approval; QR scan opens the unit record.
7. Data export, then backup and restore onto a clean environment.

15.3 Definition of done for every feature

Server-side authorisation, tenant-isolation test, audit entry, validation and error states, automated tests, documentation note, and demonstration in the review.

15.4 Defects and change control

Severity	Meaning	Effect
Critical	Wrong data, a security issue, or work blocked with no workaround	Blocks go-live
High	Major function impaired	Blocks go-live
Medium	Function affected, workaround exists	Scheduled with DSEW
Low	Cosmetic or minor	Fixed in routine maintenance

DSEW has 5 business days after each demonstration or delivery to accept it or report material deviations. A shared issue list, visible from day one, shows severity, owner and status. Anything beyond the agreed scope follows a written change request with a revised cost or schedule, approved before work (documents 1 and 2).

SECTION 16

Requirements traceability matrix

Each of the 19 modules in DSEW's enquiry is mapped to the platform capability that carries it, the evidence that proves it, and the phase in which it is delivered under Option 3 (phases as in document 2).

#	Enquiry module	Platform capabilities used	Acceptance evidence	Phase
1	CRM, enquiry, quotation, sales order	Party, workflow, approval, numbering, revisions	Linked customer → enquiry → quotation → order; revision history; approval above threshold	1
2	Project management	Work, state, milestones, budgets	Project with milestones and budget versus actual; change recorded	1
3	Engineering / PLM	Entity revisions, release gate, ECR/ECO	Released revision locked; change via ECO; order records revision	1, 2
4	Document management	Numbering, controlled workflow, files, search	Numbered document through draft to issued; distribution history; search	1
5	Manufacturing execution	Work, state machine, operations, stock ledger	Job from BOM with material allocation and operation tracking	2
6	Travellers	Templates, checkpoints, QR	Printed traveller with QR; sign-offs recorded	3
7	Quality management	Inspection, evidence, gate, NCR/CAPA	Failed mandatory check blocks release; NCR closes after CAPA verification	3
8	Barcode, QR, traceability	Batch/heat/serial, genealogy, QR identity	Forward and backward trace for a unit and a batch	3
9	Procurement	Approval, supplier, PO, receipt	RFQ → comparison → PO → receipt; approval by value	2
10	Inventory	Stock ledger, reservation, count	Balances reconcile to ledger; reversal-only correction	2
11	Planning and MRP	BOM, demand, supply, rules	Shortage recommendations from a BOM-driven plan	2
12	Testing and equipment	Test templates, instruments, calibration	Test per product family with limits; report generated; calibration alerts	3
13	Finance and accounting	Ledger, invoices, payments, projections	Invoice with tax; payment reduces balance; receivables reconcile	4
14	HR and payroll	Party, timesheets, payroll rules	Sample month reproduces DSEW's own calculation	4
15	Dispatch and logistics	Gate, packing, challan	Dispatch blocked on failed test; released after approval	3
16	Contracts and risk	Contract, obligations, alerts	Obligation and expiry alerts; LD exposure view	3
17	Dashboards and BI	Projections, permissions	Role dashboards reconcile to source records	4
18	Notifications and automation	Events, rules, scheduler	Trigger → escalation → notification demonstrated	4
19	AI use cases	Search, rules, embeddings, policy gate	Each use case meets its threshold on agreed samples	4

The MVP covers the nine modules in document 1, section 3, at the basic depth described there.

SECTION 17

Support, continuity and handover

17.1 Support during the 12-month period

Channels	team@plotarmour.in · +91 87440 69597
Hours	Business days, Monday to Saturday, 10:00 to 18:00 IST, remote
Acknowledgement	Within 1 business day
Defect plan	Fix or workaround plan for a reproducible defect within 3 business days
Included time	Up to 8 hours a month of testing assistance and maintenance
Escalation	Shubhankar Rakshit, CFO · team@plotarmour.in · +91 87440 69597
Covers	Defect warranty, testing and UAT assistance, routine maintenance (document 1, section 7)

17.2 Continuity: if PlotArmour is unavailable

- **The system runs on DSEW's server** and keeps working day to day without PlotArmour.
- **Source code** of the DSEW-specific deliverables is in a repository DSEW controls from full payment.
- **Runbooks** let DSEW's IT team, or another developer, install, back up, restore, upgrade and roll back.
- **Open components** (PostgreSQL, Keycloak, Next.js and others) are standard technology that any competent engineer can maintain, and a licence inventory is provided.
- **Data portability:** a documented export is available at handover and on request during the 12-month period.
- Updates to Verity's own components come from PlotArmour; the terms are in document 3.

17.3 Handover package

- The working, tested application and the modules of the chosen option.
- Source code of the DSEW-specific deliverables.
- Architecture overview and database schema for the delivered scope.
- Deployment and configuration instructions, and the runbooks for backup, restore, upgrade and rollback.
- Administrator and user guides.
- Document templates and the workflow configuration register.
- API notes for what the scope uses.
- Load-test report, backup and restore report, UAT sign-off records and the security check results.
- Licence inventory of open components.
- Training for administrators and users.
- Credentials through a signed handover checklist, rotated at handover.

17.4 Training

Role-based sessions for administrators, engineering, stores, production, quality, finance and management, with a quick-reference guide for the shop floor. Training is delivered remotely unless on-site work is separately agreed.

SECTION 18

Risk register

#	Risk	Impact	Mitigation	Owner
1	The host cannot run the VM at the agreed resources	Schedule, cost	Pre-deployment compatibility check first; written alternative (second VM or small dedicated host) quoted before work proceeds	PlotArmour with DSEW IT
2	Server load higher than planned	Slow screens	Resource budget, background scheduling, load test at acceptance, health view; resources increased on the same architecture	PlotArmour
3	Other workloads on the host are affected	DSEW operations	Static VM allocation, separate disks, background work windows	DSEW IT, PlotArmour
4	Document volume larger than assumed	Storage	Separate volume sized from real samples; capacity report; alerts	DSEW, PlotArmour
5	Data loss or corruption	Business continuity	Nightly backups, encryption, restored and tested twice before go-live	PlotArmour, DSEW IT
6	Certificate or DNS not ready	Schedule	Raised at kickoff; internal CA option; tested early	DSEW IT
7	Delayed inputs (samples, decisions, formats)	Schedule	Named product owner; consolidated feedback; review windows; estimates revised for pending inputs	DSEW
8	Scope growth during delivery	Schedule, cost	Signed specification with module depth; written change requests	Both
9	AI results below expectation	Value	Tiered options, acceptance thresholds on DSEW samples, AI never essential to workflow, start with search and rules	PlotArmour
10	Confidential data exposure	Reputation	Local-only AI by default, isolation, permissions, audit, NDA	Both
11	User adoption on the shop floor	Value	Phone-first screens, role-based training, quick guide, early shop-floor trials	DSEW, PlotArmour
12	Dependence on the vendor after go-live	Continuity	Source handover, runbooks, open components, export (section 17.2)	PlotArmour

SECTION 19

Glossary

Term	Meaning
PWA	Progressive Web App. A web application that can be installed on a device and opens like a native application, from one build
Tenant	The data-isolation boundary. DSEW's company data is one tenant
Row-level security	A PostgreSQL feature that makes the database itself filter every row by tenant, independent of application code
Command / Query	A request to change data, and a request to read it, handled by separate paths
Event	A recorded fact that something happened, which other modules can react to
Audit trail	The append-only record of every change
BOM	Bill of materials
ECR / ECO	Engineering change request / order
NCR / CAPA	Non-conformance report / corrective and preventive action
QAP / ITP	Quality assurance plan / inspection and test plan
MRP	Material requirements planning
RPO / RTO	Recovery point objective (data-loss window) / recovery time objective (time to restore)
vCPU	Virtual processor assigned to a virtual machine
Embedding / vector index	A numeric representation of text meaning and the index that finds similar items, used for semantic search
p95	The response time within which 95 % of requests complete
UAT	User acceptance testing
Hyper-V	Microsoft's virtualisation role in Windows Server, which hosts the Linux VM

DOCUMENT CONTROL

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